02 RAMMED EARTH
THE FRONT WALL

Privacy from the lane, rammed section by section with one small reusable form.

LEVEL	CREW	WHERE	STAGE NOW
■■□ Some skill or supervision	2–4	Front boundary, where the neighbor passes	Placement, foundation trench, sifter, form

This wall gives Hira privacy from the neighbor who walks past several times a day. It is built from free earth, sifted on site, rammed into one small form that moves along the wall piece by piece. Later, a Samarkand-style rammed-earth arch (8–10 ft) will replace the chain-link gate.

FIRST STEPS *today*

1. Wait for Yusuf's go-ahead on the setback line and height (42 in or up to 6 ft).
2. Stake and string the wall line. Leave a gap at the gate for the future arch (Yusuf sets the width).
3. Build the sifter (below) and start sifting the free earth into a covered pile. Rocks go to their own pile for the trench and plinth.
4. Make test blocks: three mixes rammed into small boxes or a cut bucket, labeled, cured two weeks, then compared.
5. Dig the foundation trench along the string line.

BEFORE YOU START

- **Height first.** In unincorporated LA County, a wall inside the front setback is limited to 42 in; outside it, up to 6 ft is generally permit-exempt. Yusuf confirms the setback line with the County before any stakes go in.
- Call 811 (DigAlert) before digging the trench.
- Ramming is jarring: wear ear protection, safety glasses and closed boots, and rotate the tamper every 20 minutes.
- Lime and cement burn skin. Gloves and glasses when mixing.
- Free fill dirt can hide ash, trash or contaminants. If a load smells, looks oily or carries debris, set it aside and ask.

THE BASICS

What it is	Damp subsoil is shoveled into a form in 6–8 in loose layers (lifts) and rammed until it rings solid. Each lift compacts by about a third. The striped layers are the look.
The mix	Aim for roughly 70% sand and gravel to 30% clay and silt. Stabilize with lime (traditional) or 6–10% cement for durability.
The drop test	Squeeze a handful into a ball. It should hold. Dropped from waist height, it should shatter into crumbs. If it smears, it's too wet; if it won't form, too dry.
Good boots, good hat	A stone plinth (local slate) keeps the base above splash and damp. An overhanging cap keeps rain off the top.
Thickness	At least 12 in; 16–18 in for a freestanding garden wall.

HOW-TO NOTES

The sifter

Frame a 3 × 5 ft rectangle from 2×4s. Screw ½ in hardware cloth across it. Prop it at about 45° on a sawhorse. Throw shovelfuls at it: fines fall through, stones roll off the front. Add a second ¼ in screen later for plaster-grade fines.

The reusable form	Two ¾ in plywood panels (form-grade), 2 ft tall × 6–8 ft long, stiffened with 2×4 walers every 2 ft. Two end boards set the wall thickness. ½ in threaded rods pass through PVC sleeves and are tightened with nuts and washers (or use long pipe clamps top and bottom). Fill and ram, strip, slide along, repeat. Next row up, stagger the joints like brickwork. One form builds the whole wall.
The tamper	An 8 × 8 in steel plate welded to a pipe handle, or a heavy 4×4 post with a plate on the end. A rented pneumatic backfill tamper is much faster.

THE PLAN *in order*

01	**Line + permit check**	Setback, height, and whether the County treats rammed earth like an ordinary wall.
02	**Earth**	Sift, store under a tarp, test blocks, choose the mix.
03	**Foundation**	Trench about 24 in wide and 18 in deep or to firm soil. Rubble trench of compacted gravel (with a perforated drain pipe sloped to daylight) or a reinforced concrete footing.
04	**Plinth**	Slate stone plinth rising about 12 in above grade.
05	**Form + first section**	Build the form, ram the first section slowly, learn the rhythm.
06	**The run**	Section by section, row by row, staggered joints.
07	**Cap**	Slate, clay tile or a lime-plastered overhang.
08	**Arch gate**	The Samarkand portal comes later as its own project.

TOOLS & MATERIALS

Shovels, mattock, wheelbarrows · ½ in and ¼ in hardware cloth · 2×4s, ¾ in plywood · ½ in threaded rod, nuts, washers, PVC sleeve · Hand or pneumatic tamper · String line, stakes, level · Tarps · Buckets for test blocks · Gloves, glasses, ear protection

FOR YUSUF TO DECIDE

- ? Wall height: 42 in or up to 6 ft
- ? Gate gap width for the arch
- ? Raw striped finish or plastered
- ? Lime or cement stabilizer
- ? Cap material
- ? Cypress planting in front of or behind the wall

Resources: LA County Public Works permits & zoning (EPIC-LA) (epicla.lacounty.gov) · DigAlert: call 811 (digalert.org)